MATH4515 Statistical and Computational Methods in Finance
Problem Set 1

***Note: The problem set is for self-practice only.**

Problem 1

We let X be a random variable and let p be the probability that X is greater or equal to a fixed number $c > 0$, i.e. $P(X > c)$.

To estimate p, a random sample of size n is drawn from the distribution of X. Given the random sample, we estimate p by

$$\hat{p}_n = \frac{Number\ of\ X_i s\ \ with\ X_i \geq c}{n}.$$

(a) Determine if $\hat{p}_n$ is unbiased estimator.
(b) Determine if $\hat{p}_n$ is consistent estimator.
(c) Calculate the mean square error of the estimator $\hat{p}_n$.

Problem 2

We let X be a random variable with the following probability density function:

$$f_X(x) = \frac{2x}{\theta^2}, \qquad 0 \leq x \leq \theta.$$

To determine θ, a random sample of size n is taken from the distribution of X. We consider the point estimator $\hat{\theta}_n = \max(X_1, X_2, \dots, X_n)$.

(a) Determine if it is unbiased estimator. If not, derive another estimator which is unbiased.
(b) Determine if $\hat{\theta}_n$ is consistent estimator.
(c) Calculate the mean square error of the estimator $\hat{\theta}_n$.

Problem 3

(a) We let N be a random variable which has Binomial distribution with parameters $n$and $p \in [0,1]$. That is, the probability mass function of N is given by

$$P(N = k) = C_k^n p^k (1-p)^{n-k}, \qquad for\ \ k = 0,1,2, \dots, n$$

Here, n is a fixed positive integer (assumed to be known).

To determine p, a random sample of size m (denoted by $\{N_1, N_2, \dots, N_m\}$) is taken.

(i) Determine the maximum likelihood estimate of p. Express your answer in terms of $N_1, N_2, \dots, N_m$.

(ii) Hence, estimate the mean square error of the MLE obtained in (a)(i).

(b) We let X be a continuous random variable which has the following probability density function:

$$f_X(x) = \frac{\theta}{x^2}, \qquad x \geq \theta,$$

where $\theta > 0$ is constant. To determine θ, a random sample of size n (denoted by $\{X_1, X_2 \dots, X_n\}$) is taken.

(i) Determine the maximum likelihood estimate of θ. (😊Hint: Express your answer in terms of $X_1, X_2, \dots, X_n$. Be careful that θ also affects the range of X.)

(ii) Hence, estimate the mean square error of the MLE obtained in (b)(i) using suitable method.

Problem 4

We let X be a random variable which has the following probability density function

$$f_X(x) = \begin{cases} \frac{(\alpha+1)x^\alpha}{\theta^{\alpha+1}} & if \;\; 0 \leq x \leq \theta \\ 0 & if \; otherwise \end{cases} .$$

Here, α and θ are positive parameters.

To estimate the parameter, a random sample of size n $\{X_1, X_2, \dots, X_n\}$ is drawn from the distribution.

(a) Suppose that θ is known, estimate α using maximum likelihood method.

(b) Suppose that α is known, estimate θ using maximum likelihood method.

(c) (Harder) Suppose that both θ and α are unknown, determine the maximum likelihood method.
(☺Hint: You can first focus on predicting the MLE of one of the parameters)

Problem 5

In this problem, we consider the daily return of a stock. The excel file "StockA.xlsx" the historical daily return of the stock from 20th Sept, 2010 to 17th June, 2019. Complete the following task:

(a) We let μ be the expected daily return of the stock. Find 98% confidence interval of μ.

(b) We let p be the probability that the stock price drops by more than 2% over a trading day. Determine if $p > 15\%$ using hypothesis test. Take the significant level to be $\alpha = 1\%$.

(c) We let σ be the standard deviation of the stock daily return. Determine if $\sigma > 3\%$. Take the significant level to be $\alpha = 5\%$.

Provide full justification to your answer.

(*To facilitate the analysis, we assume that the daily return of the stock follows normal distribution)

Problem 6

An analyst claims that a random variable X has the following probability density function:

$$f_X(x) = \frac{2}{(1+x)^3}, \qquad x > 0.$$

To test the claim, a random sample of size 10 is obtain. The sample data is presented below:

$$\{0.1,\ 0.2,\ 0.4,\ 0.5,\ 0.7,\ 0.75,\ 0.9,\ 1,\ 1.3,\ 1.6\}.$$

Suppose that the analyst would like to test the hypothesis using Kolmogorov-Smirnov test, calculate the Kolmogorov-Smirnov test statistics.

Problem 7

We adopt the setup as in Problem 6, calculate the test statistic of Anderson-Darling test.

Problem 8

We consider three stocks traded in certain market (StockA, StockB, StockC).
In this problem, we would like to examine whether the daily logreturn of three stock follows normal distribution and t-distribution by analyzing the historical returns of these stocks see the excel file "StockABC_Logreturn" which is available in canvas).
Complete the following task using *RStudio*:

- **(a)** Firstly, we investigate whether the daily return of these stock follows normal distribution.
  - **(i)** For each of the stock, plot the qq-plot against normal distribution. Give the conclusion based on the observation.
  - **(ii)** Perform the KS-test and Anderson-Darling test for each stock and give the conclusion. Here, we take the significant level to be $\alpha = 0.01$.
- **(b)** Secondly, we investigate whether the daily return of these stock follows general t-distribution.
  - **(i)** For each of the stock, plot the qq-plot against t-distribution and give the conclusion.
  - **(ii)** Perform the KS-test and Anderson-Darling test for each stock and give the conclusion. Here, we take the significant level to be $\alpha = 0.05$.